

License Decision – Campus Carbon Challenge (Sprint 1)

Project: Campus Carbon Challenge (Sprint 1 / Coursework 1)

Component covered by this decision: Backend (Python and Flask) and supporting artefacts

Databases: MySQL (Production/Runtime), SQLite (Automated Testing)

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Background and Goals

Implementing a Sprint 1 “Vertical Slice” of the Campus Carbon Challenge web application is the task for this coursework deliverable. Authentication, action logging, transparency in carbon calculations, challenge submission and scoring, moderation endpoints, and dashboards are all part of the backend. This project is primarily intended for coursework, demonstration, and future development in coursework sprints.

In order to guarantee that the test suite is quick, repeatable, and does not require an active MySQL server, the system uses MySQL in the “real/production” deployment environment and SQLite for automated testing.

What is being licensed?

The license selection is applicable to the original work developed for this project, which comprises the following:

- All of the team’s original source code including Python/Flask backend code, application logic, templates, and configuration code, if any.
- Sprint 1’s initial database schema and seed/test artifacts (such as the SQL schema and sample data created for testing and demonstration).
- Project documentation written by the team (unless otherwise specified)

The team makes no attempt to license third-party dependencies, which are subject to their own licenses.

License chosen: MIT License

The project will be licensed under the MIT License.

Rationale:

- Permissive and simple to comprehend: The MIT License has few restrictions on use, is brief, and is simple to comprehend.

- Academic reuse and iteration: It enables future developers or students to extend and reuse the codebase (for example, for research prototypes, portfolio development, or coursework continuation) while correctly attributing work.
- Compatibility: The MIT License is deemed to be compatible with most open-source software in the Python ecosystem and does not require copyleft licenses that would make reuse more complicated.

How the MIT License is applied to this project

- A 'LICENSE' file containing the full text of the MIT License should be included at the source code root (4_technical/source_code_snapshot/).
- Copyright ownership:
 - If this is group work the copyright notice should name the team or list the contributors as appropriate
- The license applies to the original code and artefacts in this repository, except where a file or directory explicitly states otherwise.

Dependencies on third-party software

Third-party libraries (such as Flask, pytest, bcrypt, cryptography, PyMySQL, and python-dotenv) are used in this project. These are installed using 'requirements.txt' and the Python packaging system; they are not copied into the repository.

Compliance strategy:

- 'software_data_inventory.xlsx' contains the license and version information for every third-party dependency.
- Dependencies are utilised in accordance with the conditions of their licenses.
- No license notices that would apply to redistributed components are removed or changed by the project. The repository mostly redistributes the team's own code because dependencies are installed independently.

Data, datasets, and factors used in calculations

To enable the system to operate during marking without the need for external data sources, Sprint 1 contains test and demonstration data.

- Synthetic seed/test data: All action logs, evidence records, seeded users, groups, and challenge submissions are made specifically for testing and demonstration purposes. There is no actual personal information included. This artificial dataset is licensed under the MIT License and is considered an artifact of the project.
- Carbon conversion factors (UK Government dataset): The UK Governments publication "Greenhouse gas reporting: conversion factors 2025" (Department for Energy Security and Net Zero) is the source of the carbon conversion factors that the backend refers to. When

reporting greenhouse gas emissions for 2025, UK and international organisations are supposed to use these factors. Unless otherwise noted, the related methodology documentation is licensed under the Open Government License v3.0 (OGL v3.0) and published as © **Crown copyright**.

How the dataset is used:

We translate user activity data into estimated CO2e impacts using a limited subset of these factors (such as factors related to food and travel). The chosen factors for Sprint 1 are seeded or stored in the application database for repeatable testing and marking. To aid in user comprehension, the application displays transparency information (factor value, unit, and explanation).

Compliance:

The dataset name, year, source, and license (OGL v3.0) are recorded in 'software_data_inventory.xlsx' and we maintain credit to the UK Government/DESNZ for the conversion factors. Our project license (MIT) only covers our original code and synthetic data; we do not assert ownership of the conversion factors dataset.

Images and Icons

Some user interface images/icons included in this project are third-party assets obtained from Google "free" resources (such as Google icon libraries). These assets are not covered by the project's MIT License. They remain under their original license terms, and the project complies by recording the source and license in 'software_data_inventory.xlsx' and providing attribution where required by the relevant license.

- We do not claim ownership of third-party icons/images.
- We will keep the original license notices and attribution.
- If any asset's licenses cannot be verified, it will be replaced with a clearly licensed alternative before final release.

Notes on operations and security (Distribution Considerations)

- The repository does not contain any secrets. Environment Variables are used to supply credentials and secrets (such as database credentials or the Flask secret key).
- Any demo credentials that are included are low-privilege accounts that are only meant to be used for demonstration and marking.
- Where evidence is uploaded, it is considered user content and is not meant for public dissemination.

Summary

To allow for easy reuse and extension while preserving credit, the project is licensed under the MIT License. Third-party software is described in 'software_data_inventory.xlsx' and is still covered by its original licenses. The seeded data is artificial and innocuous, and all images are self-made and protected by the MIT License.